

Jet polynomials and the exponential jet remainder

Claude, with Daniel Murfet

2026-09-07

Abstract

For $\xi(u) = \sum_{i < R} x_i u^i + O(u^R)$ and $g = \xi - x_0$, the u -jet of $e^{\beta s g(u)}$ to order R has coefficients $E_n(s) = \sum_{m < R} \frac{(\beta s)^m}{m!} [u^n](g_R(u)^m)$, finite sums of coefficients of powers of the Taylor polynomial g_R . We prove the exponential jet remainder $|e^{\beta s g(u)} - \sum_{n < R} E_n(s) u^n| \leq C u^R (1+s)^R e^{\beta L_1 b s}$ on $[0, b]$ from the Taylor remainder of ξ and the Lipschitz bound $|g(u)| \leq L_1 u$, using the exponential tail bound, the identity $a^m - b^m = (a-b) \sum a^i b^{m-1-i}$, and the truncation error of a polynomial on $[0, b]$. This is the first half of Astra #3's target 2b (finite compatible jets as the engine of the all-order theory). Zero sorry/axiom.

1 The things formalised

```
theorem poly_trunc_le (P : [X]) (R : ℕ) (b u : ℝ) (hu : 0 ≤ u) (hub : u ≤ b) :
  |P.eval u - ∑ n in range R, P.coeff n * u ^ n|
  ≤ u ^ R * ∑ n in Ico R (max R (P.natDegree + 1)), |P.coeff n| * b ^ (n - R)
noncomputable def jetPoly (x : ℝ → ℝ) (R : ℕ) : [X] := ∑ i in range R, C (x i) * X ^ i
noncomputable def expJet (x : ℝ → ℝ) (R : ℕ) (s : ℝ) : ℝ :=
  ∑ m in range R, (x * s) ^ m / (m.factorial : ℝ) * ((gPoly x R) ^ m).coeff n
theorem expJet_abs_le ... : |expJet x R n s| ≤ expJetConst x R n * (1 + s) ^ R
theorem exp_jet_remainder (b K L : ℝ) ... (x : ℝ → ℝ) (R : ℕ) (g : ℝ → ℝ)
  (hg : u ≤ b → Set.Icc (0 : ℝ) b, |g u - (gPoly x R).eval u| ≤ K * u ^ R)
  (hgL : u ≤ b → Set.Icc (0 : ℝ) b, |g u| ≤ L * u) (u s : ℝ) (hu : u ≤ b → Set.Icc (0 : ℝ) b)
  (hs : 0 ≤ s) :
  |Real.exp (x * s * g u) - ∑ n in range R, expJet x R n s * u ^ n|
  ≤ C (b, K, L, x, R) * u ^ R * (1 + s) ^ R * Real.exp (x * L * b * s)
```

2 Proof strategy

$e^{\beta s g} - \sum_n E_n u^n = (e^{\beta s g} - \sum_{m < R} \frac{(\beta s g)^m}{m!}) + \sum_{m < R} \frac{(\beta s g)^m}{m!} (g^m - \text{trunc}_R(g_R^m))$. The first term is the exponential tail $\leq |\beta s g|^R e^{|\beta s g|} / R!$; for the second, $|g^m - g_R^m| \leq m |g - g_R| (L_3 b)^{m-1}$ by the geometric-sum identity and $|g_R^m - \text{trunc}_R(g_R^m)| \leq u^R \text{truncConst}$ by the polynomial truncation lemma. Powers s^m , $m \leq R$, are absorbed into $(1+s)^R$.

3 Roadblocks and resolutions

- Σ is notation and cannot appear in an identifier ($\mathbf{h}\Sigma$).
- A placeholder inside `mul_comm (|_|)` leaves the type of the absolute value undetermined and stalls instance search; restructure the bound as a separate `have`.
- Distribute sums (`simp_rw [Finset.sum_mul]`) before `Finset.sum_comm`.

4 Outcome

Next: the amplitude jet $P_j(s) = \sum_{\ell \leq j} y_\ell E_{j-\ell}(s)$ with the product remainder, then the all-order one-dimensional theorem with $c_j = k^{-1} \sum_{\ell} p_{j,\ell} S_{q_j+\ell}(a)$.